ssym

October 26, 2026


```
This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
Copyright (C) 2010-2023 Joel Andersson, Joris Gillis, Moritz Diehl,
                        KU Leuven. All rights reserved.
Copyright (C) 2011-2014 Greg Horn

CasADi is free software; you can redistribute it and/or
modify it under the terms of the GNU Lesser General Public
License as published by the Free Software Foundation; either
version 3 of the License, or (at your option) any later version.

CasADi is distributed in the hope that it will be useful,
but WITHOUT ANY WARRANTY; without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the GNU
Lesser General Public License for more details.

You should have received a copy of the GNU Lesser General Public
License along with CasADi; if not, write to the Free Software
Foundation, Inc., 51 Franklin Street, Fifth Floor, Boston, MA  02110-1301  USA
```


1 SX.sym

[1]:
```
import casadi as ca
```

2 Construct using a single name

The names of the entries of the SX will be derived from the name provided as argument to SX.sym.

Without shape arguments, a 1-by-1 matrix is constructed:

[2]:
```
x = ca.SX.sym("x")
print(type(x), x)
```

```
<class 'casadi.casadi.SX'> x
```

Create a column matrix

[3]:
```
print(ca.SX.sym("x",2,1))
```

```
[x_0, x_1]
```

Create a row matrix

[4]:
```
print(ca.SX.sym("x",1,2))
```

```
[[x_0, x_1]]
```

Create a matrix

[5]:
```
print(ca.SX.sym("x",2,3))
```

```
[[x_0, x_2, x_4],
 [x_1, x_3, x_5]]
```